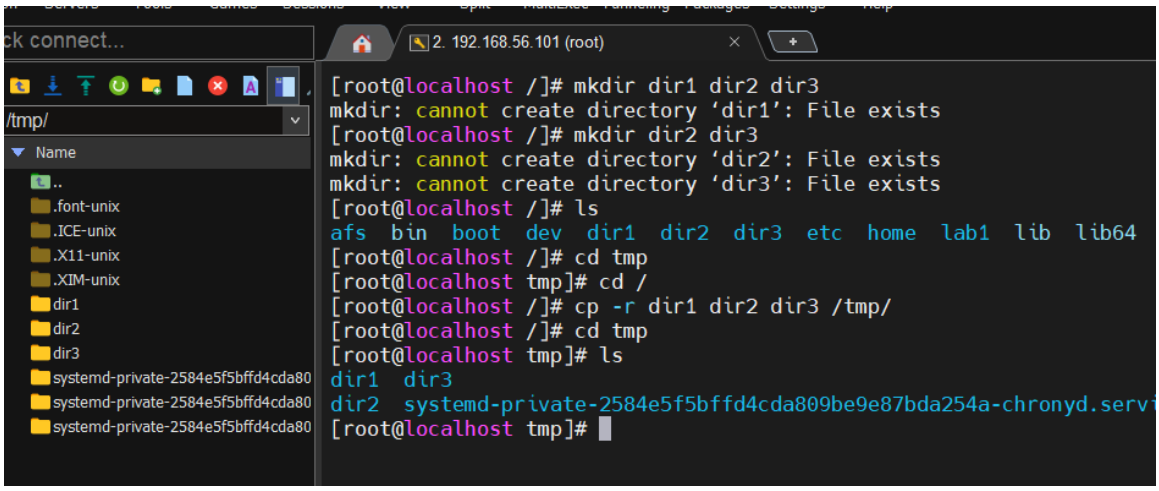
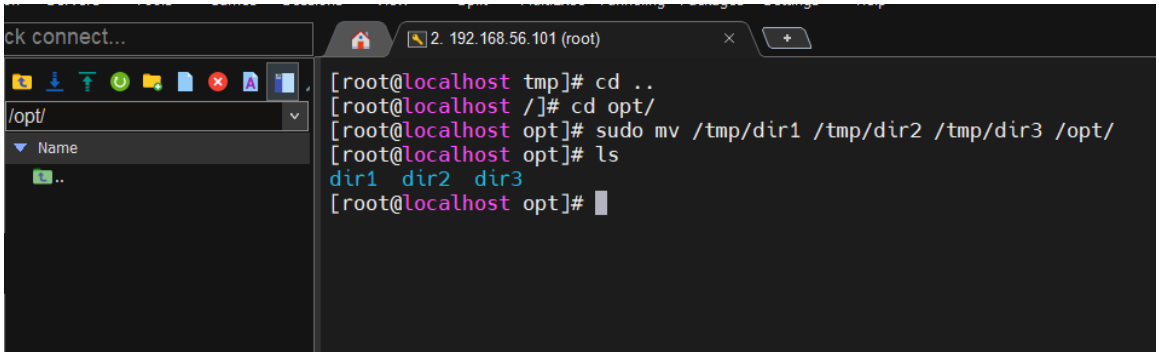
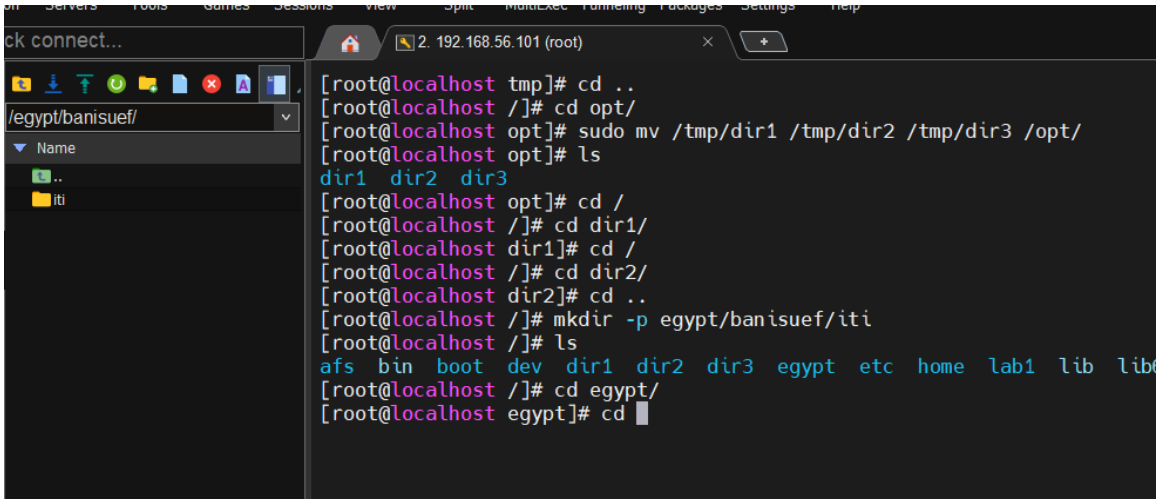
1. **Create the following directories: dir1, dir2, dir3, and copy them and their contents to /tmp. Ans:**



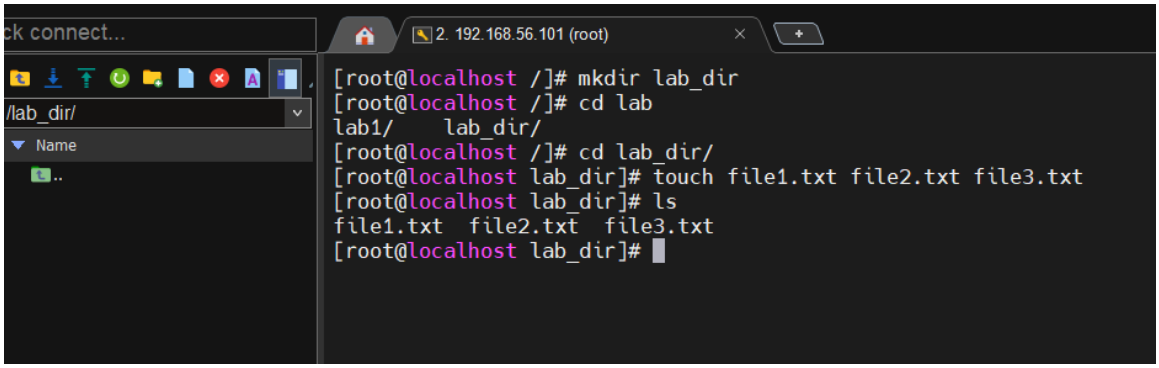
2. **Move back the three directories with their contents from the last step to /opt. Ans:**



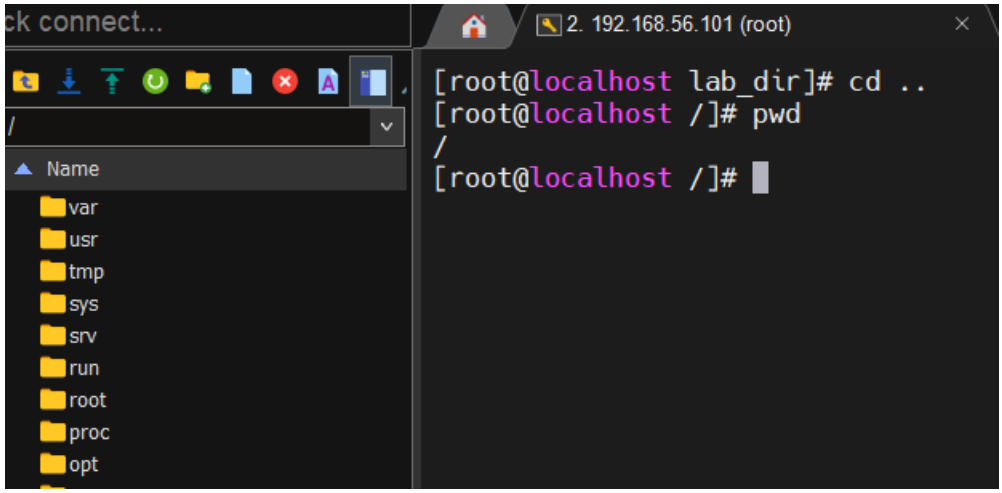
3. **Create nested directories in one command. Example: mkdir Egypt/Alex/mycompany Ans:**



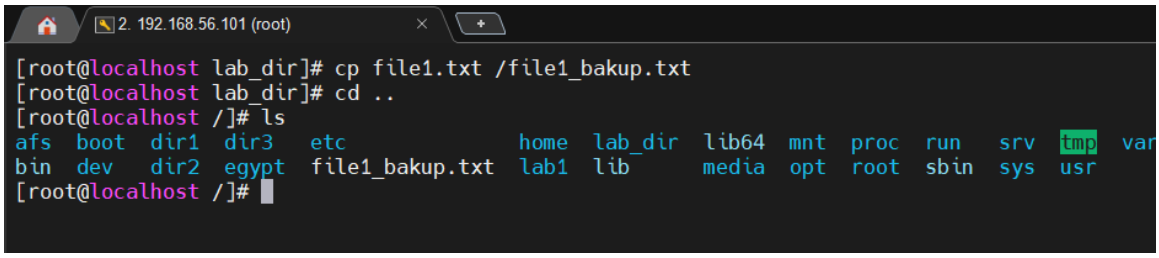
4. Create a directory named lab_dir in your home directory, and inside it, create three empty files named file1.txt, file2.txt, and file3.txt. Ans:



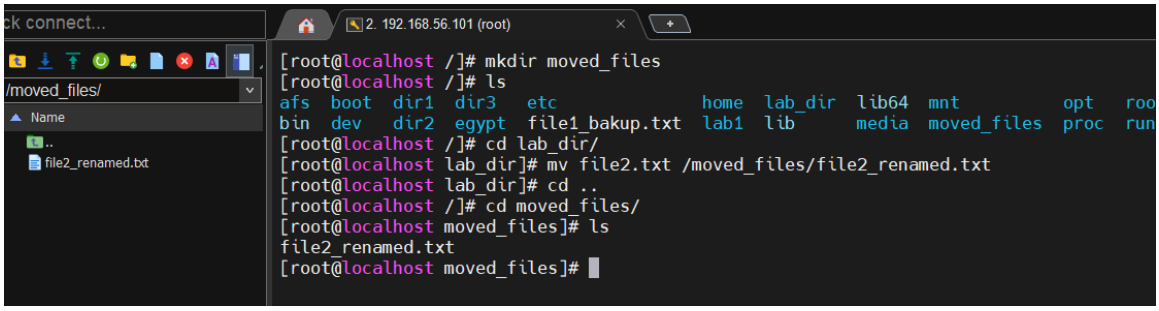
5. Go to the newly created lab_dir directory, then go back to your home directory using a relative path. Ans:



6. **Copy file1.txt to your home directory and rename it to file1_backup.txt. Ans:**



7. **Move file2.txt into a new folder called moved_files/ and rename it to file2_renamed.txt. Ans:**



8. Delete file3.txt and then delete the moved_files directory (after deleting its contents). Ans:

```
[root@localhost moved_files]# cd..
-bash: cd..: command not found
[root@localhost moved_files]# cd ..
[root@localhost /]# cd lab_dir/
[root@localhost lab_dir]# rm file3.txt
rm: remove regular empty file 'file3.txt'? y
[root@localhost lab_dir]# ls
file1.txt
[root@localhost lab_dir]# cd /
[root@localhost /]# rm -r moved_files/*
rm: remove regular empty file 'moved_files/file2_renamed.txt'? y
[root@localhost /]# cd ..
[root@localhost /]# rmdir moved_files/
[root@localhost /]# ls
afs  boot  dir1  dir3  etc              home  lab_dir  lib64  mnt  proc  run  srv  tmp  var
bin  dev   dir2  egypt  file1_bakup.txt  lab1  lib      media  opt  root  sbin  sys  usr
[root@localhost /]#
```

9. Create a text file with more than 20 lines using a loop, then use: • cat to view all contents • more and less to scroll • head to show top 5 lines • tail to show last 5 lines Ans:

```
[root@localhost lab_dir]# for i in {1..25}; do echo "this is line number $i" >> /lab_dir/lab1.txt; done
[root@localhost lab_dir]# vi lab1.txt
[root@localhost lab_dir]#
```

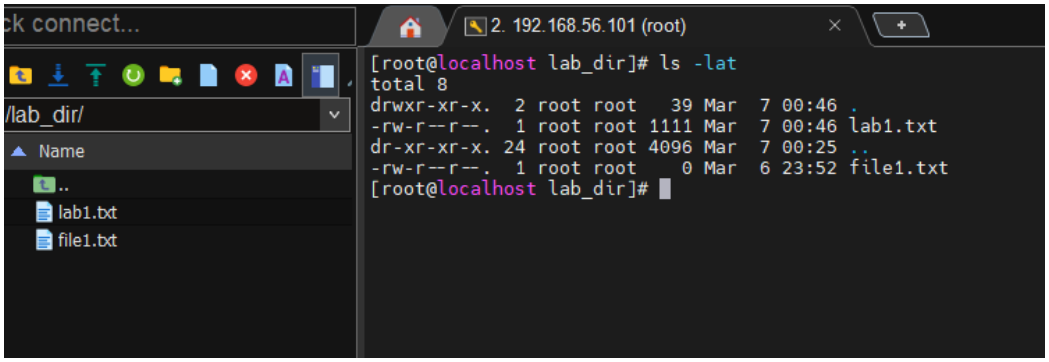
```
[root@localhost lab_dir]# cat lab1.txt

this is line number 2
this is line number 3
this is line number 4
this is line number 5
this is line number 6
this is line number 7
this is line number 8
this is line number 9
this is line number 10
this is line number 11
this is line number 12
this is line number 13
this is line number 14
this is line number 15
this is line number 16
this is line number 17
this is line number 18
this is line number 19
this is line number 20
this is line number 21
this is line number 22
this is line number 23
this is line number 24
this is line number 25
```

```
[root@localhost lab_dir]# head -n 5 lab1.txt

this is line number 2
this is line number 3
this is line number 4
this is line number 5
[root@localhost lab_dir]#
```

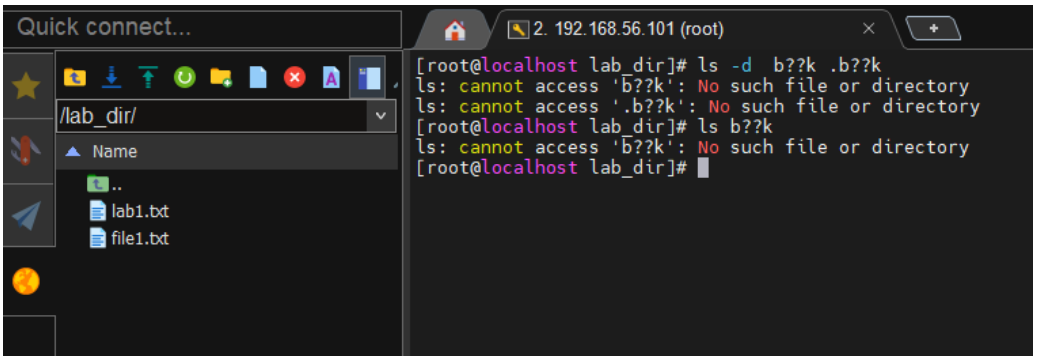
```
[root@localhost lab_dir]# more lab1.txt

this is line number 2
this is line number 3
this is line number 4
this is line number 5
this is line number 6
this is line number 7
this is line number 8
this is line number 9
this is line number 10
```

```
[root@localhost lab_dir]# tail -n 5 lab1.txt
this is line number 21
this is line number 22
this is line number 23
this is line number 24
this is line number 25
[root@localhost lab_dir]#
```

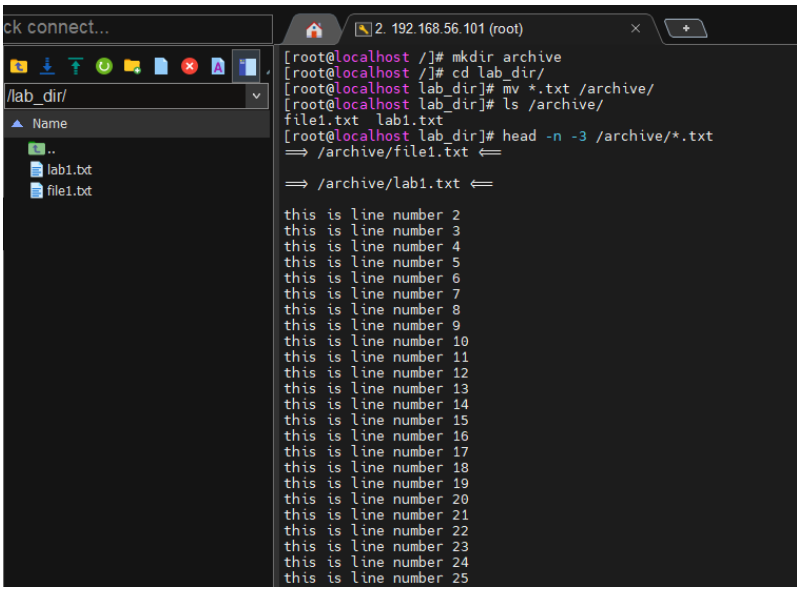
10. **List files in long format with hidden files. Then list them sorted by modification time. Ans:**



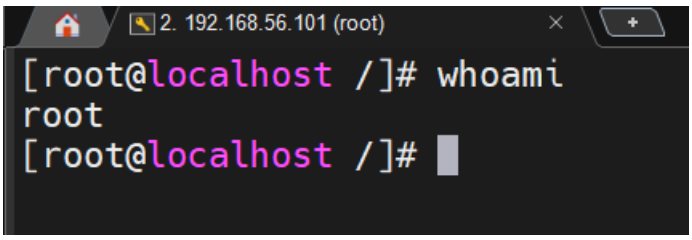
11. **List all the files/directories that are starting with "b", followed by any two characters, and ending with "k". Ans:**



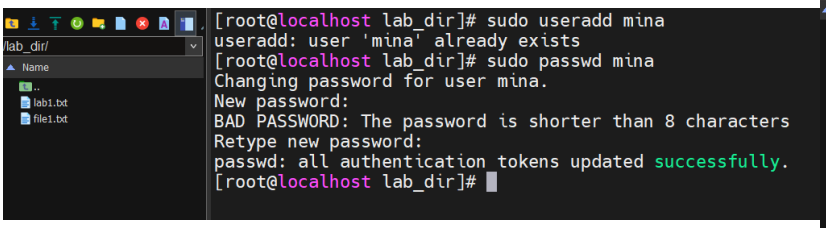
12. **Create a directory called archive, then move all .txt files into it, list the contents, and show the first 3 lines of each file. Ans:**

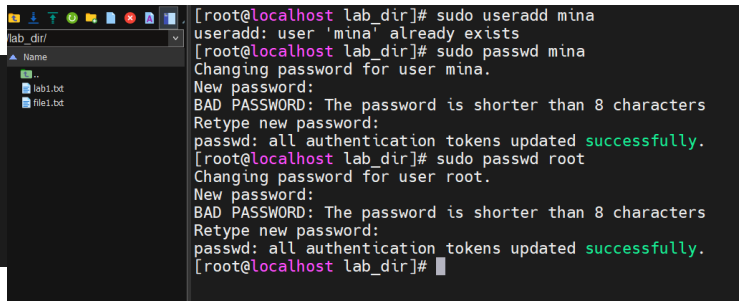


13. Know what user you are logged in with. Ans:



14. Change the password of a user and root. Ans:





15. Display all the previous commands that you have used.

Ans:

```
2. 192.168.56.101 (root)
[root@localhost /]# history
    1  ls
    2  cd /
    3  ls
    4  makdir lab1
    5  makedir lab1
    6  mkdir lab1
    7  ls
    8  ls lab1/
    9  ls
   10  cd lab
   11  cd /
   12  ls
   13  cd lab1
```

16. last users logged in Ans:

```
/root/
Name
..
.ssh
file1_bakup.txt
anaconda-ks.cfg
.tcshrc
.lesshst
.cshrc
.bashrc
.bash_profile
.bash_logout

[root@localhost ~]# last
root     pts/0        192.168.56.1     Fri Mar  6 19:59   still logged in
root     tty1                          Fri Mar  6 19:58   still logged in
reboot   system boot  5.14.0-284.11.1. Fri Mar  6 19:57   still running
root     pts/0        192.168.56.1     Fri Mar  6 17:56 - crash  (02:01)
root     tty1                          Fri Mar  6 17:46 - crash  (02:11)
reboot   system boot  5.14.0-284.11.1. Fri Mar  6 17:45   still running

wtmp begins Fri Mar  6 17:45:06 2026
[root@localhost ~]#
```